

Executive Summary

Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor

Efficient logistics and transportation systems are critical for maintaining competitiveness in modern supply chains, especially for large-scale distributors such as Nassau Candy Distributor. This project presents a data-driven analysis of factory-to-customer shipping operations with the objective of evaluating route efficiency, identifying delivery bottlenecks, and improving overall shipment performance.

The analysis is based on historical shipping records containing order dates, shipment dates, geographic routes, shipping modes, and regional classifications. A key performance metric, **Lead Time (Ship Date – Order Date)**, is used as the foundation for evaluating operational efficiency across different routes and shipping conditions.

Key Objectives

The primary objectives of this study are:

- To measure and analyze shipping lead time across different routes and regions.
- To identify high-performing and underperforming shipping routes.
- To compare efficiency across different shipping modes.
- To detect delays and bottlenecks in the distribution network.
- To provide interactive visual insights for operational decision-making.

Methodology

The dataset was processed and cleaned using Python-based data analytics tools. Missing values and invalid date entries were removed, and new derived metrics such as lead time and route combinations were created. Analytical techniques included:

- Descriptive statistics for performance measurement
- Group-by aggregation for route and regional analysis
- Distribution analysis for delay patterns
- Comparative analysis across shipping modes
- Visualization using interactive dashboards (Streamlit + Plotly)

Key Findings

The analysis revealed several important insights:

- Significant variation exists in lead time across different routes, indicating inconsistent logistics efficiency.
- Certain routes consistently outperform others, highlighting optimized distribution paths.
- Ship mode selection has a direct impact on delivery speed and reliability.
- A measurable percentage of shipments exceed acceptable delay thresholds, indicating operational inefficiencies.
- Regional disparities suggest that some geographic areas require improved logistics planning and infrastructure optimization.

Business Impact

The findings provide actionable intelligence for supply chain optimization. By identifying inefficient routes and delay-prone regions, the organization can:

- Optimize route planning and distribution strategies
- Improve shipping mode allocation
- Reduce average delivery time
- Minimize operational costs associated with delays
- Enhance customer satisfaction through faster deliveries

Conclusion

This project demonstrates the effectiveness of data analytics in improving supply chain performance. The interactive dashboard provides real-time visibility into logistics efficiency and enables data-driven decision-making. Implementing the insights from this analysis can significantly enhance operational efficiency, reduce delays, and strengthen overall distribution performance for Nassau Candy Distributor.